Branched Multi-Turn GRPO for Code-Policy Generation on a Single GPU

Abstract

We train a small instruction-tuned LLM to emit Python control policies for sandboxed RL environments using a critic-free, on-policy multi-turn RL scheme. The model interacts with the environment over an N-turn chat-template protocol whose final turn must be a committed `policy` body; intermediate turns may draft a candidate policy or run a sandboxed scratchpad. The scenario reward of the final policy supplies the only scalar signal. We replace flat trajectory-level credit assignment with a per-turn branched rollout that samples G_n continuations per node at each turn boundary and scores trajectories at the leaves. Per-turn Monte Carlo value estimates yield per-turn advantages, broadcast across the tokens of each turn — VinePPO-style credit assignment lifted from intra-completion to chat-turn granularity. To fit the resulting trajectories into a 24 GB GPU we exploit the additivity of the token-summed clipped surrogate to decompose the loss across turns, bounding peak LM-head gradient memory to $\max_n T_n$ instead of $\sum_n T_n$. We instantiate and evaluate the scheme at $N{=}2$ with $G_1{=}G_2{=}4$ on a 2D control task; the generalization to $N{>}2$ is mechanical and discussed in §6. The contribution is a clean, small-budget specialization of standard ingredients (GRPO, VinePPO, ArCHer-style hierarchical credit assignment) to the multi-turn code-policy setting, plus the per-turn loss decomposition that makes it tractable on a single L4.

1 Introduction

Reinforcement-learning fine-tuning of LLMs has converged on a small set of recipes: PPO with a learned critic, GRPO and its DAPO/CISPO variants which replace the critic with a group-relative Monte Carlo baseline, and process-supervision pipelines that train an auxiliary reward model. Most of this work targets single-turn reasoning, where the trajectory is a prompt followed by one completion and the reward is observed at the end. Extending the same recipes to multi-turn agentic settings exposes a credit-assignment problem: a single trajectory-level advantage broadcast to every model token cannot distinguish a strong first-turn strategy followed by a poor execution from the converse.

We study this problem in the concrete setting of *code-policy generation*: the LLM emits a Python `def policy(state)` body that is then executed against environment scenarios in a sandboxed subprocess. The artifact is the code; the LLM is removed from the inner control loop at deployment. The protocol uses N chat turns; intermediate turns may draft a candidate `policy` (whose scenario reward is read back as a tool message) or run a `main` scratchpad whose stdout is fed back, and the final turn commits the scored `policy`. We instantiate and evaluate the scheme at $N{=}2$.

Our training scheme has two ingredients. First, we replace trajectory-level credit assignment with a per-turn branched rollout and per-turn MC value estimates — the macro-step analog of VinePPO's intra-completion branching, with hierarchical baselines conditioned on the prefix at each turn boundary. Second, to make the resulting trajectories trainable on a 24 GB L4, we observe that the token-summed clipped surrogate is additive over disjoint token subsets sharing the same global normalizer, and exploit this to forward and backward each turn independently.

Peak LM-head gradient memory drops from $O(B \cdot T_{\text{full}} \cdot |V|)$ to $O(B \cdot \max_n T_n \cdot |V|)$ at gradient cost zero; the parameter gradients summed across the per-turn passes equal the single-pass gradient exactly.

The combination is a natural specialization of standard mechanisms rather than a new algorithm; the contribution is a clean assembly that fits on a single small GPU and the loss-decomposition trick that makes it possible.

2 Related work

Critic-free LLM RL. GRPO [1] and its token-summed variants (DAPO [2], CISPO) replace PPO's learned critic with a group-relative MC baseline. We use the token-summed form because its additivity over token subsets is what makes our per-turn loss decomposition exact.

Branching for credit assignment. The closest existing method is VinePPO [3], which estimates $\widehat{V}(s_t)$ at chosen intra-completion states via on-policy MC re-branching and uses it as a per-token baseline. We use the same mechanism but branch only at *turn boundaries* of a chat trajectory and broadcast the resulting advantage as a constant within each turn — group-relative within a turn, VinePPO-like across turns.

Hierarchical multi-turn agents. ArCHer [6] adopts the same macro-step decomposition (per-turn value, per-token policy update) but learns the per-turn value with a critic. We estimate it by MC over G_2 continuations, trading variance for the cost of fitting a value head. ReBN [5] replaces GAE with batch-normalized MC returns; the hierarchical mean we use can be viewed as ReBN partitioned by turn-1 prefix. RAGEN's StarPO [4] is the canonical flat-baseline critic-free MC scheme for multi-turn LLM RL; setting $G_2 = 1$ collapses our scheme to it.

$G_1 \times G_2$ tree rollouts. The same tree shape appears in process-reward-model training pipelines such as Math-Shepherd [7] and OmegaPRM [8], where the per-prefix MC mean labels intermediate steps for a learned reward model. We use the tree as a policy-gradient baseline rather than as a labelling oracle.

Code policies. Voyager [9] and FunSearch [10] also produce executable code as the artifact, but rely on frozen-LLM prompting plus skill-library accumulation or evolutionary mutation respectively. We adapt the LLM by gradient-based RL of LoRA weights and add a scratchpad turn between prompt and committed code.

3 Method

3.1 Setup

The agent is a language model π_θ over chat-template trajectories. The base is `Qwen2.5-Coder-3B-Instruct` with frozen weights W; θ parameterizes a LoRA adapter $W' = W + AB^\top$. A task $\mathcal{T}$ specifies environment source code, S scenarios $\{\sigma_i\}$ that vary the seed and hidden parameters, and a per-scenario reward. The scoring oracle runs a candidate policy body in a sandboxed subprocess against each scenario and returns $R(\text{code}; \mathcal{T}) = \sum_i r_i(\text{code}; \sigma_i)$. Validation failures, errors, and timeouts on σ_i score the *error floor* $r_i = -200$, calibrated below any running-but-failing trajectory — otherwise errored rollouts ($r = 0$) outrank running-but-poor rollouts and the gradient rewards crashing.

3.2 Chat-template protocol

Each turn the model emits one fenced code block tagged `policy` or `main`. `policy` defines `def policy(state)`, which the tool executes across all scenarios and reports the summed reward as a user-role message. `main` is arbitrary scratchpad code with no environment access; the tool returns truncated stdout. The final turn must be `policy`; `main` or untagged final turns are floored. Malformed intermediate turns are likewise floored and surfaced as a textual rejection. A length-N trajectory is $\tau = (s, y^{(1)}, m^{(1)}, \ldots, y^{(N-1)}, m^{(N-1)}, y^{(N)})$, with completions $y^{(n)}$ alternating with tool messages $m^{(n)}$, and $R(\tau) = R(\text{code in } y^{(N)})$ the only scalar reward. The remainder of the paper specializes to N=2: a single intermediate turn (draft policy or scratchpad) followed by a committed `policy`. Generalizing the rollout, advantages, and loss decomposition below to N>2 is mechanical (see §6).

3.3 Token-summed clipped surrogate

For trajectory b at position t, with importance ratio $\rho_t^b = \pi_\theta/q$ and per-token advantage A_t^b,

$$\mathcal{L}(\theta) = -\frac{1}{N_{\text{tok}}} \sum_{b,t} \mathbf{1}\{\text{model token}\} \cdot \min\big(\rho_t^b A_t^b,\ \text{clip}(\rho_t^b,\ 1-\epsilon_{\text{lo}},\ 1+\epsilon_{\text{hi}}) A_t^b\big), \tag{1}$$

with N_{tok} the count of model tokens in the global batch.[1] (1) is *additive over disjoint token subsets*: for any partition of model-token positions, the loss equals the sum of partial losses computed with the same N_{tok}. This is the property the per-turn decomposition (§3.6) exploits.

3.4 Branched rollout

For each prompt s, sample G_1 turn-1 variants in one `model.generate` call: $y_i^{(1)} \sim q(\cdot \mid s)$, $i = 1, \ldots, G_1$. For each $y_i^{(1)}$ execute the tool to obtain m_i, then sample G_2 continuations $y_{i,j}^{(2)} \sim q(\cdot \mid s, y_i^{(1)}, m_i)$ in one call. Score each leaf to obtain $\{R_{i,j}\}$. Generate-call cost per prompt is $1 + G_1$, versus $2G_1G_2$ for an unbranched rollout with the same number of leaves.

3.5 Per-turn MC value estimates and advantages

Two empirical values, both pure averages over the tree:

$$\widehat{V}_0 = \frac{1}{G_1 G_2} \sum_{i,j} R_{i,j}, \qquad \widehat{V}_1^{(i)} = \frac{1}{G_2} \sum_{j} R_{i,j}. \tag{2}$$

$\widehat{V}_0$ estimates the initial-state value; $\widehat{V}_1^{(i)}$ the value after committing to turn-1 variant $y_i^{(1)}$. Per-turn advantages, broadcast to every token of the corresponding turn:

$$A_i^{(1)} = \widehat{V}_1^{(i)} - \widehat{V}_0 \qquad \text{(over } y_i^{(1)}\text{)}, \tag{3}$$

$$A_{i,j}^{(2)} = R_{i,j} - \widehat{V}_1^{(i)} \qquad \text{(over } y_{i,j}^{(2)}\text{)}. \tag{4}$$

The turn-2 advantage uses the conditional mean given the turn-1 prefix rather than the global per-prompt mean: a poor turn-2 sample drawn from a strong turn-1 is recognized as poor instead of inheriting its prefix's above-average outcome. Tool-message tokens are masked via `tool_mask`; their advantage is zero.

[1] This matches the DAPO and CISPO recipes; per-row averaging variants (original GRPO, BNPO, Dr.GRPO) divide by per-row token counts.

3.6 Per-turn loss decomposition

The token-summed clipped surrogate (1) factors over any partition of the model-token positions into disjoint subsets that share the same global normalizer N_{tok}. Letting $\mathcal{I}_n$ index the model tokens of turn n,

$$\mathcal{L} = \sum_{n=1}^{N} \mathcal{L}_{\mathcal{I}_n}, \qquad N_{\text{tok}} = \sum_{n=1}^{N} |\mathcal{I}_n|. \tag{5}$$

Per-row averaged variants (original GRPO, BNPO, Dr.GRPO) do *not* factor this way: each row carries its own denominator, so the partial sums do not combine into the single-row loss. The token-summed form is what makes the decomposition exact.

We exploit (5) to compute $\mathcal{L}$ in N forward/backward passes rather than one. Pass n forwards $[s \,|\, y^{(1)} \,|\, m^{(1)} \,|\, \ldots \,|\, y^{(n)}]$, computes $\mathcal{L}_{\mathcal{I}_n}$, backwards, and frees. By (5) the parameter gradients accumulated across the N passes equal the single-pass gradient exactly.

Why this saves memory. The dominant activation cost is the LM-head logit tensor of shape $(B, T, |V|)$, with $|V| \approx 152{,}000$ for our base model. A single forward over the full trajectory materializes logits at every position from the start of $y^{(1)}$ through the end of $y^{(N)}$ — the model tokens of the non-final turns sit in the middle of the sequence, so they cannot be reached by the standard trailing-window optimization (`logits_to_keep`, see Appendix A). The per-turn forwards each have $y^{(n)}$ as a trailing block, so the per-pass logit memory is bounded by $\max_n T_n^{\max}$ instead of $T_1 + \sum T_{m_n} + \cdots + T_N$. Concrete numbers for N=2 appear in Appendix A; at our settings the saving over the smallest single-forward variant is roughly 2×.

Restrictions. Beyond the token-summed normalizer required by (5), the implementation also needs $\beta = 0$ (a KL term to π_{ref} would require per-segment slicing of reference log-probabilities) and `num_iterations`= 1 (so cached `old_per_token_logps` need no re-segmenting).

4 Experimental setup

Environment. A 2D puck-on-tilted-plane controller. The puck moves in $[-6, 6]^2$ under a constant hidden tilt (t_x, t_y) and linear drag μ; the policy receives $\{p_x, p_y, v_x, v_y, g_x, g_y\}$ and must drive the puck to a goal within 120 steps. Hidden parameters — tilt, drag, thrust limit $a_{\max}$, initial velocity — vary per scenario and are not in the state. The thrust envelope is set to $a_{\max} = \|\text{tilt}\|_2 \times 1.05 + \epsilon$ so a naive straight-at-goal policy fails because it cannot cancel tilt drift. Per-step reward is $-\Delta t \cdot \text{dist}(p, g)$ with a +50 bonus on reaching the goal; errored episodes hit the floor of §3.1.

Baseline gap. Two hand-written policies fix the headroom: naive aim (≈ 26% goal rate over 200 seeds) and PD-with-tilt-feedforward (≈ 73%). The gap is the room LoRA fine-tuning has to exploit.

Model and compute. `Qwen2.5-Coder-3B-Instruct`, LoRA rank $r = 16$ on attention and MLP projections, on a single 24 GB L4 GPU. bf16 weights, gradient checkpointing on. We use $G_1 = G_2 = 4$ (16 leaves per prompt) and a per-turn token cap of 400.

Training. The token-summed clipped surrogate (1) with $\epsilon_{\text{lo}} = 0.2$, asymmetric upper clip, $\beta = 0$, and the per-turn advantages of (3)–(4). No reference KL, no critic, no GAE, no context truncation across turns.

Diagnostics. We log $\widehat{V}_0$ and the between-variant standard deviation of $\widehat{V}_1$, the within-variant standard deviation, and their ratio (the *signal ratio*); the per-turn advantage standard deviations; the fraction of trajectories at the error floor; and the count of `generate` calls per prompt (expected $1 + G_1$). The signal ratio guides allocation between G_1 and G_2 at fixed budget: > 1 means turn-1 variation dominates and a larger G_1 helps; < 1 means within-variant noise dominates and a larger G_2 helps.

5 Evaluating the per-turn credit assignment

The per-turn loss decomposition (§3.6) is purely a memory enabler — it does not change the gradient. The substantive claim that needs empirical support is the *credit-assignment* contribution: that hierarchical per-turn advantages (3)–(4) on a $G_1 \times G_2$ tree do better than a single trajectory-level advantage broadcast across both turns. We propose a six-arm ablation (Table 1), ordered roughly from the weakest baseline to our method. Adjacent rows isolate a single axis where possible (E vs. D isolates the tree; F vs. E isolates the per-turn baseline); the frozen-to-trained jump (B vs. C) and the single-to-multi-turn jump (C vs. D) are the two places where two axes change at once, and we read those rows accordingly.

Arm	Rollout structure	Advantage	What it isolates over the previous arm
A — Frozen, one-shot	N=1 inference, no training	—	Base model's prior on the code-policy interface
B — Frozen, multi-turn prompted	N=2 inference, no training	—	Does the two-turn protocol help at inference time? (Voyager-/FunSearch-style frozen-LLM use)
C — Single-turn GRPO	G flat N=1 rollouts, trained	Standard GRPO advantage	Does RL fine-tuning help at all?
D — Flat StarPO	G independent N=2 trajectories, trained	$A(\tau) = R(\tau) - \bar{R}$, broadcast to all tokens	Does multi-turn training help over single-turn? (RAGEN-style)
E — Branched, flat	$G_1 \times G_2$ tree, trained	$A_{i,j} = R_{i,j} - \bar{R}$ on all tokens	Does the tree alone help (more leaves per generate call), independent of credit?
F — Branched, hierarchical (ours)	$G_1 \times G_2$ tree, trained	$A_i^{(1)} = \widehat{V}_1^{(i)} - \widehat{V}_0$ on $y_i^{(1)}$; $A_{i,j}^{(2)} = R_{i,j} - \widehat{V}_1^{(i)}$ on $y_{i,j}^{(2)}$	The per-turn baseline contribution

Table 1: Ablation arms, ordered from weakest baseline (top) to ours (bottom). Trained arms (C–F) match leaf count $G = G_1 G_2 = 16$ so they see the same number of scored trajectories per step. D uses $2G$ `generate` calls per prompt; F uses $1 + G_1$. Report both equal-rollouts and equal-wall-clock comparisons. Frozen arms (A, B) use the same Qwen2.5-Coder-3B-Instruct base with no gradient updates, and serve as training-free references analogous to Voyager- or FunSearch-style frozen-LLM uses of the code-policy interface.

Primary metric. Held-out goal rate on a stress distribution (more extreme tilts/drag than train) at the best checkpoint, reported as median and IQR over ≥ 3 seeds per arm, plotted against gradient steps and against wall-clock. Statistical test: paired Wilcoxon over per-seed best-checkpoint goal rates; report effect size, not just p. The minimum convincing comparison is F > D at matched leaves.

Diagnostic checks. Two further measurements sharpen the mechanism story.

1. *Turn-1 quality curve.* Track the mean reward of turn-1 `policy` drafts (when turn-1 is in policy mode) as a function of training step. Under D the trajectory-level advantage projects onto turn-1 only through the global mean; under F, $A_i^{(1)}$ provides a direct gradient signal on turn-1 quality. F should show a steeper turn-1 reward rise than D.
2. *Signal-ratio conditioning.* The training loop already logs the per-prompt signal ratio $\widehat{\sigma}(\widehat{V}_1^{(i)})/\overline{\widehat{\sigma}(R_{i,\cdot})}$. Bucket prompts by their iteration-0 signal ratio. Ratio > 1 means turn-1 choice dominates the variance and per-turn credit should help; ratio < 1 means within-variant noise dominates and per-turn credit should not. F's advantage over D should grow with the bucket's signal ratio. If it does not, the mechanism is not doing what theory predicts and the headline claim weakens.

Failure modes to declare upfront. If F $\not>$ D under matched leaves, the hierarchical baseline does not pay off in this domain and the contribution reduces to "branching amortizes generate calls" (E vs D). If F $\approx$ E but both beat D, the gain is from the tree structure, not from the per-turn baseline. If the trained arms (C–F) do not clearly beat the frozen arms (A, B), RL fine-tuning is not buying anything over the base model's prior under our protocol, and the right framing for this work is prompt-engineering, not training. If B > A, the multi-turn protocol itself helps even without training, and that gap should be reported as part of the headline. Either outcome is informative; we report whichever obtains.

6 Limitations

We instantiate and evaluate the scheme at N=2. The generalization to N>2 is mechanical: each $\widehat{V}_n^{(\cdot)}$ becomes the conditional MC mean over the subtree below it, and the two-pass loss decomposition becomes an N-pass decomposition by the same additivity argument ((5)). Neither is conceptually new, but both add bookkeeping and we have not implemented or benchmarked them here. We assume deterministic environments given the seed — under stochastic dynamics the leaf rewards become noisier estimates of state values and G_2 would have to grow accordingly. The $\beta = 0$, `num_iterations`= 1 restrictions are implementation choices that the same decomposition argument can lift with extra per-segment slicing of reference log-probabilities and cached sampling log-probabilities.

7 Conclusion

We described a critic-free, on-policy multi-turn RL scheme for code-policy generation that combines branched rollouts with per-turn MC value estimates and a per-turn loss decomposition. The algorithmic ingredients are individually standard; the assembly is small enough to train on a single 24 GB GPU, and the per-turn decomposition — which the token-summed normalizer makes exact — is what keeps it that way as the number of turns grows.

A Memory accounting and the trailing-window LM-head

This appendix walks through why a single forward over a multi-turn trajectory is expensive and where the per-turn decomposition's saving comes from. It is written for readers familiar with policy-gradient RL but not necessarily with the internals of LLM training. The argument turns on a single API constraint in HF `transformers` which we make explicit here rather than in the body.

The LLM as an autoregressive policy. A causal LLM is a stochastic policy whose action space at each step is the vocabulary $\mathcal{V} = \{1, \ldots, |V|\}$. Given a sequence of T token IDs as input,

the model produces, at each position t, a logit vector $\ell_t \in \mathbb{R}^{|V|}$; the next-token distribution is $\text{softmax}(\ell_t)$, and the log-prob of having emitted token a at position $t{+}1$ is $\log \text{softmax}(\ell_t)_a$. To form the policy-gradient surrogate (1) we need exactly these log-probs at every position t such that $t{+}1$ was a *model-emitted* token. Logits at other positions — prompt positions, and (in our multi-turn setting) tool-message positions — are not needed for the gradient and are wasted compute if we materialize them.

Where the memory goes. Internally the model has two pieces. The *transformer body* is a stack of blocks (attention + MLP) that turns input tokens into hidden states of shape (B, T, d) with $d \approx 2048$ for Qwen-3B; this is where the bulk of the $3{\cdot}10^9$ parameters live. The *LM head* is a single final linear layer of shape $(d, |V|)$ that projects each hidden state to its logit vector. The LM head has only $d \cdot |V| \approx 3{\cdot}10^8$ parameters — modest. But its *output activation* $(B, T, |V|)$ is roughly $|V|/d \approx 75\times$ larger than any per-position activation in the body, and the autograd graph saves an equal-sized buffer for backward. At our trajectory lengths these LM-head tensors are the only thing that does not fit on a 24 GB L4; the transformer body is comfortably under budget.

The `logits_to_keep` parameter. Because most positions' logits are not actually needed for the gradient, HF `transformers` lets you skip them. The forward call accepts an integer argument `logits_to_keep` (formerly `num_logits_to_keep`); setting it to K replaces the full projection `lm_head`(hidden) with `lm_head`(hidden$[:, -K{:}, :]$). The LM head is applied only to the *last* K hidden states and the output logits have shape $(B, K, |V|)$ instead of $(B, T, |V|)$. The transformer body still runs over all T positions because causal self-attention chains every position to its prefix — but the body is two orders of magnitude cheaper per position than the LM head, so this is a clean win.

The constraint that matters. K counts *trailing* positions. There is no API to ask for logits at, say, the set of positions $\{50, 51, 200, 201\}$ — only at "the last K." To get logits at a non-trailing block of positions, you must either enlarge the trailing window to cover it (paying for every position in between, including ones whose logits you don't need) or run a separate forward whose sequence ends at the block.

Why multi-turn is expensive in a single forward. In our $N{=}2$ trajectory $[s \,|\, y^{(1)} \,|\, m \,|\, y^{(2)}]$, the model tokens of $y^{(2)}$ are trailing but those of $y^{(1)}$ are not. The smallest trailing window covering both blocks runs from the start of $y^{(1)}$ through the end of $y^{(2)}$ — $T_1 + T_m + T_2$ positions. The tool-message tokens contribute nothing to the loss (they are masked) but their logits are still materialized because they sit inside the window. As N grows, the gap widens: the single-forward window becomes $T_1 + T_{m_1} + T_2 + \cdots + T_N$, the per-turn-forward bound stays at $\max_n T_n$.

Three regimes. Concretely, with $T_1 = T_2 = 400$, prompt ≈ 500, tool message ≈ 50, batch $B = 16$ leaves per prompt, $|V| = 152{,}000$, and 4-byte logits (the autograd intermediate is held in fp32 for the cross-entropy reduction even when weights are bf16):

The honest comparison for our setting is (b) vs. (c) — a roughly $2\times$ saving here, growing with T_m and with N. Regime (a) is shown for completeness but is not what a careful single-forward implementation actually does.

What the per-turn decomposition is and isn't. It is not a new memory optimization over `logits_to_keep`; it is the natural way to apply that optimization when model tokens span multiple non-trailing blocks. The non-trivial observation is that the gradients accumulated across per-turn passes equal the single-pass gradient *exactly* (eq. 5), and that this exactness depends on

Regime	Positions through LM head	Logit tensor
(a) Single pass, no `logits_to_keep`	$T_{\text{full}} = 1350$	≈ 13.1 GB
(b) Single pass, `logits_to_keep` $= T_1{+}T_m{+}T_2$	850	≈ 8.3 GB
(c) Two passes (ours), `logits_to_keep` $= \max(T_1, T_2)$	400 per pass, freed between	≈ 3.9 GB

Table 2: LM-head logit tensor size under each regime, plus an equal-sized autograd buffer for the backward pass (not shown). Regime (a) is the truly naive forward. Regime (b) is what an HF-fluent implementation does by default; the trailing window has to span both turns because `logits_to_keep` only trims leading positions. Regime (c) is the per-turn decomposition of §3.6.

the token-summed normalizer of (1). Per-row-averaged GRPO variants would have to fall back to a single forward (regime b) or accept a normalizer approximation.

Implementation note. Pass 1 forwards $[s \mid y^{(1)}]$ as a self-contained sequence with `logits_to_keep` $= T_1$. Pass 2 forwards the full trajectory but packs $[s \mid y^{(1)} \mid m]$ into the prompt slot of HF's training interface and $y^{(2)}$ alone into the completion slot, so `logits_to_keep` $= T_2$ behaves as expected.

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